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Sales Modeling With Economic Indicators

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Data Summary

	A	B	C	D	E	F	G	H	I	J	K	L
1		product_ic	location_ic	state	departme	_201405	_201406	_201407	_201408	_201409	_201410	_201411
2	0	7	516	NY	Clinique	0	0	0	0	0	0	0
3	1	10	353	NY	Kitchen El	1106	287.04	1334.88	692.16	406	1845.6	146.32
4	2	10	516	NY	Kitchen El	70	33.28	75.6	56	85.84	148.8	57.04
5	3	14	8	NJ	Clinique	26	45.76	86.4	38.08	238.96	93.6	47.12
6	4	14	10	NY	Clinique	800	468	503.28	665.28	450.08	444	133.92
7	5	14	32	CA	Clinique	62	85.28	51.84	1460.48	452.4	540	262.88
8	6	14	57	FL	Clinique	98	62.4	62.64	67.2	90.48	36	47.12
9	7	14	88	CA	Clinique	918	47.84	185.76	2943.36	403.68	314.4	32.24
10	8	14	141	CA	Clinique	322	322.4	142.56	174.72	53.36	249.6	86.8

- 999 product-location rows
- 155 weeks (Feb 2014–Jan 2017)
- Total cells = 154,845

- **Initial Cleaning Steps:**

- Remove trailing-year zero → 77 rows deleted
- Nullify leading pre-launch zeros → 14517 cells out of 154,845(155×999) = 9.3%

- **Anomaly Detection Approaches:**

Method	What it does	Result
Modified Z-Score	Detects trend outliers	14803 cells flagged (10.3%)
Seasonality-aware residual	Detects seasonal deviations	0 cells flagged
Rolling % Change	Guards against sudden shifts	72 clean rows($k=1$, 265 $k=3$, 554; $k=5$, 875)

- **Aggregation Options:**

- By Product (298 rows) – used for model
- By Location (321 rows) – not used due to data sparsity

- **Reasons:**

- Product-level patterns = cleaner trends
- Location-level data = too sparse/erratic

Data Augmentation

- **Gaussian Noise:** Add random variation to data
- Purpose: Improve generalization and reduce overfitting
- **Accuracy:** Top10 nMAE 0.31-0.43 → 0.25-0.34

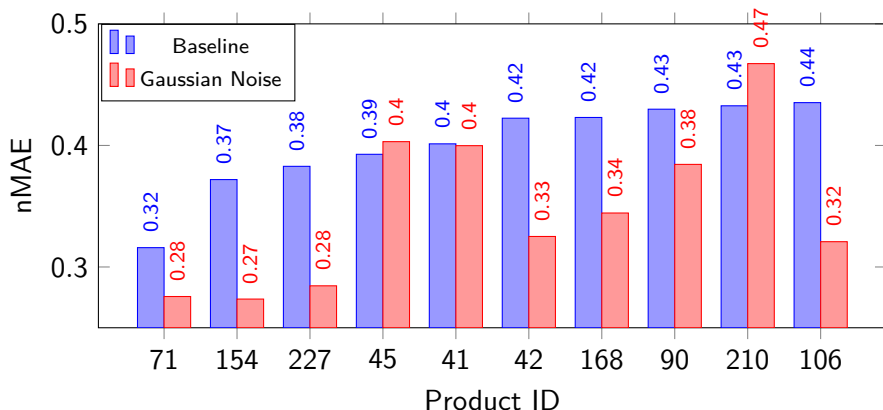


Figure: nMAE Comparison of Top 10 Products: Baseline vs. Gaussian Noise

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 - Great for structured data. Easy to train and tune.

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- Feed forward network
 - Type of neural network where information flows in one direction, from input to output. No cycles or feedback loops

LSTM/XG Boost Parameters

- LSTM
 - Window size of 24, hidden size of 16, 1 layer, and 100 epochs
- XG Boost
 - N estimators is 500, learning rate of 0.01, and max depth of 9
- Baseline Performances (Product)
 - XG Boost Model: 0.275 nMAE
 - LSTM: 0.316 nMAE

Feature	Importance
353_lag_1	0.135890
1429_rolling_std_5	0.127123
1263_rolling_mean_3	0.100379
623_lag_1	0.070093
752_rolling_std_5	0.069776
1436_momentum	0.053268
566_momentum	0.035335
997_momentum	0.034656
1008_momentum	0.034055
961_rolling_mean_3	0.032660

Measuring Accuracy

- Definitions

- Our forecast predictions are represented by $F = (f_1, f_2, \dots, f_n)$
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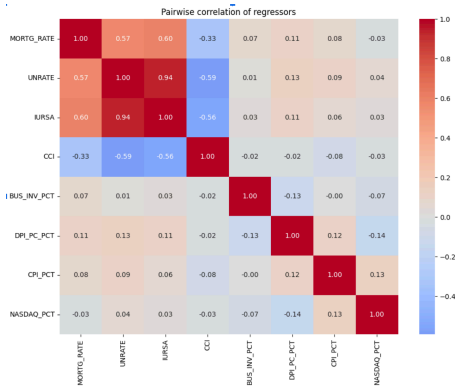
- **Normalized MAE**

- Normalized MAE = $\frac{\frac{1}{n} \sum_{i=1}^n |f_i - v_i|}{\frac{1}{n} \sum_{i=1}^n |v_i|}$
- Allows for fair comparison across products

Multicollinearity Check(1)

- Pairwise Pearson Correlation Heatmap

- High-correlation pairs ($|r| \geq 0.8$): UNRATE IURSA $\rightarrow r = +0.9377$



Multicollinearity Check(2)

- **Variance Inflation Factor (VIF)**

Variable	VIF
UNRATE	8.962530
IURSA	8.916796
MORTG_RATE	1.584676
CCI	1.550863
DPI_PC_PCT	1.086566
NASDAQ_PCT	1.057646
CPI_PCT	1.054268
BUS_INV_PCT	1.033329

Table: Rule of Thumb for VIF Interpretation

VIF Range	Interpretation	Action
$VIF \leq 5$	Low/moderate overlap	Safe
$5 < VIF \leq 10$	Noticeable	Consider thinning
$VIF > 10$	Severe multicollinearity	Drop or combine

Multicollinearity Check(3)

- **Condition Number of the correlation matrix:**
- $7.0 < 30 - 35 \rightarrow$ the whole regressor block is numerically well-conditioned.
- **Conclusion:**
 - Most regressors are safe to use
 - But IURSA and UNRATE carry near-duplicate information - additional testing to choose which one to retain

Regressor Results

- Economic Indicators had both positive and negative effects.
- IURSA, CCI, DPI, and NASDAQ closing prices all improved the regression model.
- CPI, Gas Prices, and Average Business Inventory all created similar performance as the baseline.

	Regressor	NMAE	MAPE	CVRMSE	
1	Baseline	0.32-0.44	24.67-42.56	93.41-1037.18	
2	Gaussian	0.25-0.34	21.61-34.31	81.15-533.59	
3	DPI	0.24-0.37	26.22-56.09	42.868-265.08	
4	Gas Prices	0.30-0.41	31.31-54.27	49.21-983.84	
5	Business_inventories	0.32-0.41	30.54-56.13	66.61-844.35	
6	CCI	0.23-0.33	25-28	62-320	
7	CPI	0.32-0.40	27.53-46.80	57.82-967.14	
8	NASDAQ	0.29-0.35	25.68-36.65	56.98-396.89	
9	IURSA	0.24-0.4	24.72-51.21	47-249	

Multiple Regressors

- CCI and NASDAQ closing yielded worse performance
- CCI, IURSA, and DPI together (our 3 helpful regressors) led to better performance than the baseline

	Regressor	NMAE	MAPE	CVRMSE
10	CPI and NASDAQ	0.34-0.41	24.69-45.75	41.48-404.90
11	CPI and IURSA and DPI	0.24-0.36	25.31-53.56	42.67-249.63
12	CCI and IURSA and Gaussian	0.25-0.36	25.06-43.49	41.47-248.83

Fourier Norms Definition

- Let $f: \mathbb{Z}_N \rightarrow \mathbb{R}$ and $\widehat{f}$ be its fourier transform

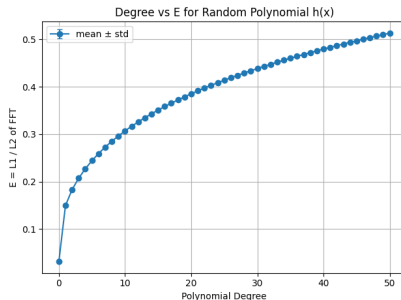
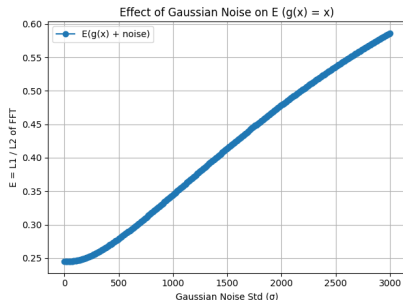
$$\|\widehat{f}\|_{L^1(\mu)} = \frac{1}{N} \sum_{m \in \mathbb{Z}_N} |\widehat{f}(m)|$$

$$\|\widehat{f}\|_{L^2(\mu)} = \left[\frac{1}{N} \sum_{m \in \mathbb{Z}_N} |\widehat{f}(m)|^2 \right]^{1/2}$$

$$\mathcal{E} = \frac{\|\widehat{f}\|_{L^1(\mu)}}{\|\widehat{f}\|_{L^2(\mu)}}, \quad \frac{1}{\sqrt{N}} \leq \mathcal{E} \leq 1$$

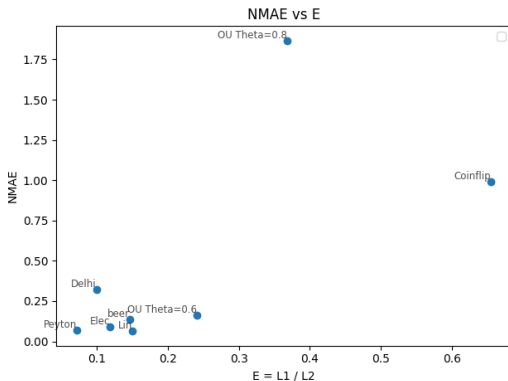
- Low $\mathcal{E}$: spectrum concentrated in few frequencies $\rightarrow$ well-approximated by a low-degree trigonometric polynomial $\rightarrow$ more forecastable
- High $\mathcal{E}$: spectrum spread across many frequencies $\rightarrow$ noise-like $\rightarrow$ less forecastable

Synthetic Data: $\mathcal{E}$ Rises with Noise and Complexity



- Noise $\uparrow \Rightarrow \mathcal{E} \uparrow$ (approaches random-sequence level)
- Polynomial degree $\uparrow \Rightarrow \mathcal{E} \uparrow$ (complexity acts like noise)

Real-World Data: $\mathcal{E}$ Predicts Forecast Error



- Clear positive relationship between $\mathcal{E}$ and NMAE
- Higher $\mathcal{E} \Rightarrow$ higher forecast error
- Correlation: Pearson $r = 0.8577$ ($p = 0.0289$), Spearman $\rho = 0.6571$ ($p = 0.156$)

L_1 Imputation

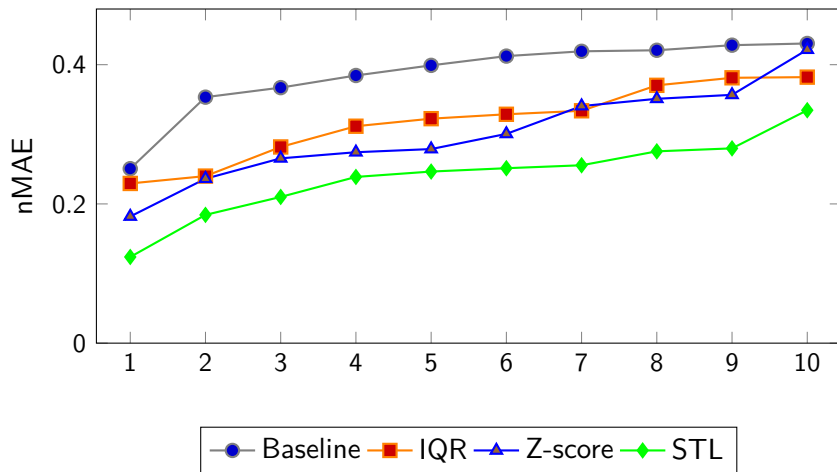


Figure: Top-10 nMAE by method (ranked) – Line Plot.

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- Extend the analysis to multivariate settings with $f: \mathbb{Z}_N \rightarrow \mathbb{R}^n$
- Examine sensitivity of $\mathcal{E}$ to sampling rate, seasonality, and preprocessing choices